

Computer Programming Assignment 2

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Question 1: Array Stats

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];
    int sum = 0;

    cout << "Enter 10 numbers: " << endl;
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];                // taking input from user
        sum += arr[i];                // keeping track of total
    }

    int big = arr[0];                  // assume first is biggest
    int small = arr[0];                // assume first is smallest

    for(int i = 1; i < 10; i++) {
        if(arr[i] > big) big = arr[i]; // update if found bigger
        if(arr[i] < small) small = arr[i]; // update if found smaller
    }

    cout << "Max: " << big << endl;    // showing max
    cout << "Min: " << small << endl;  // showing min
    cout << "Avg: " << (float)sum / 10 << endl; // average calculation
    return 0;
}
```

Question 2: Array Reversal

```
#include <iostream>
using namespace std;

void flip(int a[], int n) {
    int start = 0, end = n - 1;
    while(start < end) {
        int temp = a[start];          // temporary storage
```

```

        a[start] = a[end];           // swapping start and end
        a[end] = temp;              // finishing the swap
        start++;                     // move forward
        end--;                       // move backward
    }
}

int main() {
    int data[] = {1, 2, 3, 4, 5};
    flip(data, 5);                  // calling reverse function
    return 0;
}

```

Question 3: Matrix Sum

```

#include <iostream>
using namespace std;

int main() {
    int a[3][3], b[3][3], res[3][3];

    for(int i=0; i<3; i++)
        for(int j=0; j<3; j++) cin >> a[i][j]; // fill first matrix

    for(int i=0; i<3; i++)
        for(int j=0; j<3; j++) cin >> b[i][j]; // fill second matrix

    for(int i=0; i<3; i++) {
        for(int j=0; j<3; j++) {
            res[i][j] = a[i][j] + b[i][j]; // add both matrices
            cout << res[i][j] << " ";    // print result
        }
        cout << endl;                    // row break
    }
    return 0;
}

```

Question 4: Student Name Sorting

```

#include <iostream>
#include <string>
using namespace std;

int main() {
    string n[5];
    for(int i=0; i<5; i++) cin >> n[i]; // inputting names

    for(int i=0; i<4; i++) {
        for(int j=i+1; j<5; j++) {

```

}

Question 3. Power Load Analysis

}

Question 6: Class Topper

}

Question 7: Renewable Energy Monitoring

```
#include <iostream>
using namespace std;

int main() {
    float solar[24];
    float s = 0;
    for(int i=0; i<24; i++) {
        cin >> solar[i];           // get solar data
        s += solar[i];             // add to total
    }
    cout << "Average: " << s / 24 << endl; // daily average
    return 0;
}
```

Question 8: Battery Health

```
#include <iostream>
using namespace std;

void checkV(float v[]) {
    int bad = 0;
    for(int i=0; i<20; i++) {
        if(v[i] < 3.0 || v[i] > 4.2) bad++; // check safety limits
    }
    if(bad > 0) cout << "Warning!" << endl; // alert user
}

int main() {
    float bats[20];
    checkV(bats); // monitor voltages
    return 0;
}
```

Question 9: Smart Grid

```
#include <iostream>
using namespace std;

int main() {
    float grid[4][7];
    for(int i=0; i<4; i++) {
        float rTot = 0;
        for(int j=0; j<7; j++) {
            rTot += grid[i][j]; // sum for each region
        }
    }
}
```

```
        cout << "Region " << i+1 << ": " << rTot; // output result
    }
    return 0;
}
```

Question 10: Attendance System

```
#include <iostream>
using namespace std;

int main() {
    int rolls[5][7];
    int p = 0;
    for(int i=0; i<5; i++)
        for(int j=0; j<7; j++) p += rolls[i][j]; // count all present
    cout << "Total present: " << p << endl;      // show total
    return 0;
}
```